

Active ants: dynamics

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?

• **Part 1 (Rakotomalala & W, in review, 2025):**

o Existence of compact global attractor

■ Robinson, *Infinite-Dimensional Dynamical Systems*

9 Attractor has dimension at least 4 if and only if it is linearly unstable



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 - Attractor has dimensional lower bound of 4 if homogeneous state is linearly unstable
 - Linearised operator around $f = 1/2\pi, L$ (**not** self/skew-adjoint)

